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Dashboarding: Where Data Becomes Insights

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Contents

Abstract **2**

Part 1: Getting Started with Your Report **3**

Data Dictionary *3*

Creating a Report and Accessing Data *5*

Exploring the Data – and Finding Stuff to Fix! *6*

Part 2: Turning Data into Information **9**

When No-Shows Happen *9*

Pricing Analysis *10*

Automatic Explanations *11*

How No-Show Factors Affect Profit *12*

Part 3: Love it – But Make it Interactive **14**

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Abstract

This hands-on workshop focuses on uncovering the story hidden in your data through effective dashboarding in SAS® Visual Analytics. You'll explore a real-world airline scenario and learn how to build meaningful visualizations. We'll see how thoughtful visual design can help answer practical questions, such as understanding ticket pricing behavior or identifying factors that contribute to flight no-shows. Step by step, we'll create graphics that surface patterns and trends, transforming data from static numbers into a story that supports better decisions.

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Part 1: Getting Started with Your Report

Data Dictionary

The AltitudeExpress dataset provides a comprehensive view of flight-related information, including customer demographics, flight details, and ticket pricing. It can be used to analyze patterns such as no-show rates, the impact of weather, and the relationship between ticket prices and flight distances. The dataset is simulated, with one observation recorded per customer. During the analysis, additional variables will be created to supplement the report.

The table contains 23 columns and 10,000 rows. The variables used for this session are:

Column Name	Description
Customer_ID	Unique identifier for each customer
No_Show	Indicates whether the customer showed up for the flight (0=Yes, 1=No)
Major_Hub_Departure	Indicates if the departure was from a major hub (0=No, 1=Yes)
Departure_Day	Day of the week when the flight departed
Departure_Hour	Time range of the departure
Connections	Number of connections during the flight
International_Flight	Indicates if the flight was international (0=No, 1=Yes)
Connection_International_Flight	Indicates if any connection was international (0=No, 1=Yes)
Business_Traveler	Indicates if the customer is a business traveler (0=No, 1=Yes)
Booking_Class	Class of booking (e.g., Coach, Business, First)
When_Ticket_Purchased	Time frame when the ticket was purchased
Age_Grouping	Age group of the customer
Frequent_Flier_Status	Indicates if the customer is a frequent flier (0=No, 1=Yes)
Weather_Issues	Indicates if there were weather issues during the flight (0=No, 1=Yes)
Weather_Issues_Connection	Indicates if there were weather issues during any connection (0=No, 1=Yes)

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Gender	Gender of the customer
Exact_Departure_Date	Exact date of departure (January 1, 2025-May 1, 2025)
Flight_Distance	Distance of the flight (e.g., Short-haul, Medium-haul, Long-haul)
Flight_Delay_Minutes	Number of minutes the flight was delayed
Seat_Type	Type of seat (e.g., Window, Aisle, Middle)
Ticket_Price_USD	Price of the ticket in USD
Exact_Ages	Exact age of the customer
Departure_Day_Num	Numerical representation of the departure day

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Creating a Report and Accessing Data

You've just joined AltitudeExpress, a dynamic airline company known for its innovative approach to data-driven decision-making. Your mission is to dive into the company's flight data and uncover insights that can help optimize operations and enhance customer satisfaction.

Equipped with a wealth of data, your task is to explore the data and create compelling visualizations that reveal hidden patterns and trends. This will not only help AltitudeExpress minimize flight no-shows but also improve ticket pricing strategies and overall efficiency.

1. Navigate to the upper left corner in SAS Viya Landing, click **Applications** > **Explore and Visualize**. This takes you to Visual Analytics.

   SAS Visual Analytics is a web-based product that leverages SAS High-Performance Analytics technologies to empower organizations to explore huge volumes of data very quickly to identify patterns, trends, and opportunities for further analysis.

2. Select **New report** from the top right corner.

3. Expand the **Data** pane on the left and click **Add data**.

4. Using the search, find the **ALTITUDEEXPRESS** dataset.

5. Browse through the **Columns** in the ALTITUDEEXPRESS dataset to better understand the data, then **Add** it to your report.

6. Now that we've loaded the dataset into our report, we see the variables in the **Data** pane.

   Notice that next to the **Category** variables, there is a number indicating the cardinality.

   Notice that next to **Measure** variables, there are:

   - Outlier icons that indicate outliers have been detected for that variable
   - Privacy icons that indicate the variable might contain private information that might identify an individual.

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Exploring the Data – and Finding Stuff to Fix!

The *No_Show* variable is going to be of interest to us because we want to minimize the number of people who miss their flights. This is inefficient and if we could have predicted it, we could oversell tickets to ensure flights are full.

1. To investigate *No_Show*, drag it from the **Data pane** onto the canvas. You will see a "+ Auto chart" indicator meaning that SAS is going to choose the best visualization given the variable you want to analyze.

   A histogram is created! We can see there are significantly more *No_Shows* between 0 and 1 than between 1 and 2. But… we also know that the *No_Show* variable is either equal to 0 (No) or 1 (Yes) from the data dictionary. So, it would be better if we could analyze these as distinct values.

2. Duplicate the *No_Show* variable by right-clicking on the variable in the **Data pane** and then selecting **Duplicate.**

3. Click on **Edit properties** to the right of the newly created *No_Show (1)* variable, and change the **Classification** from Measure to Category.

4. Now you have two variables for *No_Show*; one that is a measure and one that is a category. Notice that the categorical variable has 2 distinct values. Change the name of each of the variables to *No_Show_Num* for the numeric version, and *No_Show* for the categorical variable.

9. Feel free to explore other variables by dragging them onto the canvas. You can place these items anywhere on the canvas - a blue rectangle will appear to indicate where the object will be dropped.

   For instance, dragging *Exact_Ages* onto the canvas creates a histogram showing the frequency and variability.

5. Wanna get fancy? Select both *Booking_Class* and *Ticket_Price_USD* then drag them onto the canvas.

   A bar chart is created! Notice that the scale for the *Ticket_Price_USD* is in millions – this is because ticket prices are being aggregated, and the default aggregation is sum. This chart is likely not very representative of individual ticket prices since First class is more luxurious and often more expensive than Business or Coach.

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6. You can change the aggregation method of *Ticket_Price_USD* by, you guessed it, using the **Edit properties** next to the variable and changing the **Aggregation** to **Average.**

   Check out the bar chart! We are now looking at average ticket prices by booking class and as expected, the highest average ticket price is for first class tickets.

7. Whew! No data is ever perfect, so we'll need to perform some additional data prep before we move onto our analysis.

   There are several other variables that should be defined as categories because they represent a binary variable, where 0=No and 1=Yes. Select the following variables:
   a. *Business_Traveler*
   b. *Connection_International_Flight*
   c. *Connections*
   d. *Frequent_Flier_Status*
   e. *International_Flight*
   f. *Major_Hub_Departure*
   g. *Weather_Issues*
   h. *Weather_Issues_Connection*

   Right-click one of the variables and select **Convert to categories**.

8. Now for aggregations! Ensure you do not have any other variables selected – you can use the *Clear selection* option at the bottom of the variable list – then choose:
   a. *Exact_Ages*
   b. *Flight_Delay_Minutes*

   Right-click on either of the variables and choose **Aggregation** > **Average**.

9. For the next variable, find *Exact_Departure_Date* in the Category section. SAS has a special way of working with dates; however, *Exact_Departure_Date* was not loaded with the proper format for SAS to automatically treat it as such. This will prevent the variable from being used in date-specific roles. We can fix that, though!

   At the top of the Data pane, select **+ New data item** > **Calculated item**

10. Name the variable *Flight_Date*. Using the **Functions** drop-down, choose the **Parse** function in the **Text (simple)** section.

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11. The Parse function will be added, and the right-side menu will appear. Make the following selections:

    Value: *Exact_Departure_Date*
    Format: use the pencil icon to choose ANYDTDTE (the first option). This format will take the values of *Exact_Departure_Date* and convert them to date values.

    A preview of your results will appear at the bottom of the window. Select OK when you're done reviewing.

12. Now that we've converted, modified, and added new variables, we can remove some of the redundant or unnecessary variables. Select the following:
    - *Exact_Departure_Date*
    - *Age_Grouping*
    - *Customer_ID*
    - *Departure_Day_Num*

    Right-click and select Hide.

13. Look at all the great work you've done! Now is a good time to save, so find the Save button in the upper right corner of the report.

14. Select My Folder and give your report a meaningful name.

You're all set to start your analysis!!

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Part 2: Turning Data into Information

When No-Shows Happen

1. Add a new page to your report by clicking the **+** button next to Page 1.

2. Select *No_Show_Num* and *Departure_Hour* from the **Data pane** and drag them onto the canvas.

   Looks like the greatest number of no-shows happen overnight (6:00 pm - 6:00 am), followed closely by the morning hours (6:00 am – 12:00 pm).

3. What about no-shows by day of the week? Duplicate the *No_Show_Num by Departure_Hour* bar chart by clicking the options button in the upper right corner of the histogram and selecting **Duplicate**.

4. Seeing double? No worries – we'll make a quick switch in the **Roles** pane on the right sidebar. Click *Departure_Hour* to **Replace Data Item** with *Departure_Day*.

   And voila! We can now see when no-shows happen throughout the week. Looks like Mondays have the highest number of no-shows, likely due to challenges associated with transitioning from weekend to weekday, such as rescheduled plans or post-weekend fatigue. You may also notice the default is to sort in descending *No_Show_Num* order. We want to create our own sort order to put the days of the week in order (Monday, Tuesday, …, Sunday).

5. On the **Data** pane, right-click *Departure_Day* and choose **Custom sort.**

6. Double-click each day of the week (in order) to move it to the **Sorted Items** box. Click **OK**.

7. On the *No_Show_Num* by *Departure_Day* bar chart, right-click within the graph and select **Sort > Departure_Day: Ascending**.

   Much better! Our view now shows how no-shows fluctuate throughout the week.

10. Let's check if there's any patterns with no-shows over time. To view this, simply drag *Flight_Date* and *No_Show_Num* onto the canvas.

    There are some peaks and valleys; however, nothing is jumping out.

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11. To further investigate, expand the **Data Roles** pane on the right side. There are many different roles for a Time Series plot. Let's take advantage of the grouping option by selecting **+ Add** next to the **Group** role, then select *When_Ticket_Purchased*.

    Your time series plot is now grouped and color coded by purchased category. Looks like most no-shows happen amongst those who purchase their ticket same day!

12. This page is looking good! A more descriptive page name will help an external audience know what this page is all about before clicking on it. So, rename your current **Page 2** by clicking the **Options** menu (three dots) and selecting **Rename**. Give your page a helpful name, such as "When No-Shows Happen".

13. Before going any further, don't forget to save !

Pricing Analysis

1. Add another page and rename it "Pricing Analysis" where we will dive into where we can make some $$$!

2. You have already created a bar chart with details about ticket prices and booking class, so let's add that to your page.

   Go to **Page 1** and locate the *Ticket_Price_USD by Booking_Class* bar chart. Click the options button in the upper right corner of the chart. Hover over **Move to** and select your new **Pricing Analysis** page.

3. We've seen the price breakdown by booking class – but what about the overall ticket pricing? Drag *Ticket_Price_USD* onto the *Pricing Analysis* page.

   From this histogram, we see that ticket prices are skewed right, with the majority of tickets being sold for under $400. There is a group of tickets priced between $400-$600 and only a handful of tickets priced around $900.

4. Another chart that's great at showing variability is a box plot! On the left sidebar, expand the **Objects** pane and double-click the **Box plot** object to add it to your page.

5. Click **Assign data** and add *Flight_Distance* to the **Category** role and *Ticket_Price_USD* to the **Measures** role. Finally, click **Apply** then **Close**.

   A box plot showing the distribution of ticket prices is generated for each flight distance category. As we might expect, longer flights cost more, while shorter flights are cheaper.

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6. We could benefit from a few additional options to highlight the variability. Expand the **Options** pane on the right sidebar. Under the **Box Plot** section, make the following adjustments:
    - Outliers: **Show outliers**
    - Check box for **Averages**

    Outliers will be shown using shaded bins and averages will be denoted by a square. We can now see that all averages lie above the median, sometimes even above the 3rd quartile mark, and there are several outliers for all flight distances. The box and whiskers are relatively small, indicating that the prices don't vary a lot within the flight distance groups.

Automatic Explanations

1. Let's turn our attention back to no-shows. This time we'll focus on the category version of the variable. From the **Data pane**, right-click the categorical *No_Show* variable, hover over **Explain** then select **Explain on new page**.

    A new page is created with an **Automated Explanation** object, which determines the most important underlying factors for a specific response variable, *No_Show* in our case.

2. Rename this page to **Automated Explanation.**

3. Change the **Event level** to **1** in the upper right corner since we want to analyze the factors that caused the customer to miss their flight.

4. Click on the different factors to see their relationship with *No_Show*.

    Clicking on the *Connections* factor reveals that although there are significantly fewer cases where a flier has a connection (Connections=1), it appears there is an almost equal number of no-shows for the two groups. In other words, the percentage of no-shows is different for the two groups. Could this be right?!

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How No-Show Factors Affect Profit

1. To calculate the percentage of no-shows in each group, we can create a new calculated variable! Right-click *No_Show_Num* from the **Data pane** and select **New calculation**.

2. Under **Type,** select **Percent of total – Sum**. Feel free to rename the variable if your heart so desires!

3. Now for the breakdown! Create a new page called "How No-Show Factors Affect Profit."

4. Drag your newly created *No_Show_Num (Percent of total – Sum)* variable – or whatever you named it – and the *Connections* variable onto the canvas.

   We were right, there's a higher percentage of people who didn't show up for their flight when they had a connection.

5. Let's utilize a new type of object, the bubble plot, which is particularly effective for visualizing relationships between multiple variables by using several dimensions of data. To uncover how no-show factors impact profit, we will use our newly created variable *No_Show_Num (Percent of total – Sum)*, together with the two factors most strongly associated with no-shows—*Flight_Delay_Minutes* and *Connections*—as well as *Ticket_Price_USD*.

   From the **Objects** pane, find **Bubble plot** under **Graphs** and drag the object onto the canvas.

6. Click **Assign data** and select the following roles, then click **Close**.
   - X axis: *Ticket_Price_USD*
   - Y axis: *Flight_Delay_Minutes*
   - Size: *No_Show_Num (Percent of total – Sum)*
   - Group: *Connections*

   As we've learned, the no-show percentage is larger when a passenger has a connection, but also the flight delay is longer and the price is higher. Pay attention to the scale though, because these differences aren't as large as they appear.

7. One more role we might want to incorporate into our analysis is the **Animation**. We can use this to see how our measures change over time, helping us uncover discrepancies or anomalies in the data and enabling users to drill down into specific days of interest for deeper investigation. Add *Flight_Date* to the **Animation** role for the bubble plot.

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The bubble plot is updated to show the evolving, day-by-day relationship between the variables displayed. You can either click the play button to see the measures move through time, use the forward and back arrows to step through, or use the slider to navigate to a particular day.

8. At this point, we've built out a lot of great visualizations! There's always more that can be done to polish your dashboard, such as standardizing colors, adding chart titles, adjusting placement within a page, etc.; however, with a few small changes, we can enhance this report for our end user!

   Take a moment to save your report before moving on!

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Part 3: Love it – But Make it Interactive

1. Head back to the "When No-Shows Happen" page and expand the **Actions** pane on the right sidebar. Click the box to select **Automatic actions on all objects**.

   With one easy click, you can now select a bar for day of the week and see the other graphs on the page filter to that selection. You could also start in the *No_Show_Num by Departure_Hour* bar chart to filter by time. Use this opportunity to click around and test the interactivity! When you're finished, you can clear your filter choice at the top of the page.

2. Feelin' fancy? Make the same **Automatic actions on all objects** selection on the **Pricing Analysis** page.

Although your time turning data into information is coming to a close for today, there are endless possibilities to explore; the adventure of turning data into wisdom is just beginning.

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